



DNESC 4211 Programming For Analytics

Spring 2026

Instructor

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The views expressed in this course are mine personally, and they do not necessarily reflect the views of the US Department of Treasury or the US Government.

Class Logistics

Funger 209

Tuesday, Thursday

4:45 PM - 6:00 PM

Office Hours: Funger 320 at 6:10PM Thursdays (10 minutes after class)

Textbooks and Materials

Instructor Provided. Weekly materials will be uploaded via Blackboard or Google Drive/Colab

Recommended, but not required materials. These have been helpful to previous students

- Python for Data Analysis: Data wrangling with Pandas, NumPy, and IPython by Wes McKinney, O'Reilly Media, 2012, ISBN 978-1-4493-1979-3
- An Introduction to Statistical Learning: with Applications in R (Springer Texts in Statistics) 2nd ed. 2021 Edition
- An Introduction to Statistical Learning: with Applications in Python (Springer Texts in Statistics) 1st ed. 2023 Edition

Course Overview

Coding is a tool in your tool kit to rapidly explore, manipulate, experiment and eventually derive insights from data. It allows you to interact with larger amounts of data and explore faster, but it is useless without analytic ability or ability to communicate value and findings to your stakeholders. This course is structured to reflect the importance of all three skills, analytic ability, communication and coding, and you will need all three to be successful. Early in the course you will select a dataset corresponding to a business sector or industry that interests you. You will explore this dataset using your new coding skills throughout the entire semester, delivering insights and findings along the way. This course will effectively be structured as an MBA course, where the presentation of the results, not just the results themselves will make you fully successful.

Expected Time Investment

This class will likely require more time outside class than your average course. The reason for this is that programming is really learned by doing, and thus it will be expected that students spend time completing exercises, homeworks, Datacamp, and Semester Long Project to maximize the exposure to computer programming. A general rule of thumb is that students should expect to spend about six hours a week on assignments, independent learning, and Semester Long Project a week, in accordance with [University credit hour policy](#).

Learning Outcomes

- Exposure to R and Python tools and techniques used to solve realistic business problems
- Create and interpret data visualizations using R & Python
- Apply statistical modeling techniques for descriptive and predictive analytics
- Communicate results and insights from data analysis for the level of a business stakeholder

A Note on Plagiarism

There is a plethora of materials regarding to Python and R on the web, which is something that makes them very useful and special. They both have a large community of users out there that are ready and willing to help. Students should leverage posts on sites like StackExchange, though need to be careful not to plagiarize from them. Students should use these resources to help them solve a problem, but should not copy directly from these posts.

- Do not google the answers to questions...
- Do google how to achieve certain procedures...

For example, Given the prompt: "Please find all prime numbers between 3 and 100":

- Do **not** google "How to find prime numbers in Python."

- Do google “How to test if a number is evenly divisible by another in Python”
- Do google “What is a prime number”
- Do google “Python syntax for a for-loop”

AI

This is further complicated with Generative AI's strong ability to produce code. Students should use Generative AI in a similar manner as stated above. You should also remember that there is a big difference in the code produced from generative AI compared with that of a student that is first learning how to code, and it is quite obvious where students use it.

The overall rule of thumb is that you are here to learn coding. If you can use AI in a way that assists you, then please do so. But do not let it do the work for you. Experienced programmers use AI all the time, but they also know how code works so they can better guide and check the AI code development.

Last note, exams are hand written, so you'll need to know the basics that we go over in the lectures.

Tech Environments

We will primarily use Google Colab for in-class work. Students are free to use any Python/R distribution and text editor for their personal work. It is recommended that students make use of a public facing Github to demonstrate their work. This can be very useful for resumes and job interviews. Students should also download Anaconda locally to make use of Rstudio, Spyder, Jupyter etc locally. Your local machines will have better processing power than what is offered in the base Google Colab environment and will be useful for Semester Long Projects.

In your colab environment, or whatever environment you are using, I suggest you turn off AI capabilities in settings. (See above on AI.)

Late Policy for Assignments

All assignments lose 5 points a day for being late. No assignments will be accepted more than 7 days late.

Semester-Long Project

Through the course, students will find a dataset for a particular industry or sector that interests them, or ideally, one where they would like to one day work to give them the opportunity to analyze a dataset of interest on their own terms. Through the course, students will apply their new-found analytic skills with R and Python to their dataset and create 4 business deliverables (Power Point, Word Docs etc) for a semi-technical stakeholder. The final work product, and possibly others will be delivered by the student orally to a small group of students. This work

can be done in either Python or R. Students will create the following four work products:

- Semester-Long Project Deliverable #1: Students will have picked an industry, data set and defined their stakeholder (for whom the products will be geared). Student will have clearly laid out what the value of the data will bring to their stakeholder. This work product should be viewed as a “pitch to the stakeholder” to invest company time into exploring and using the dataset.
- Semester-Long Project Deliverable #2: Students will summarize the data, define the limitations of the data, and present some initial insights to the stakeholder to show that their work has value for the business.
- Semester-Long Project Deliverable #3: Students will apply supervised and/or unsupervised algorithms to the dataset and create a work product explaining it.
- Semester-Long Project Deliverable #4: Students will wrap up the semester with an oral delivery of the comprehensive project. This is an opportunity for students to take on feedback that was given to them along the way and adjust to meet client demands.

Prerequisite

Students need to ensure that they successfully complete the Math/stat requirement. Some prior programming experience (e.g., a high school programming course) in any programming language is strongly encouraged. This is not an introductory programming course.

Attendance

Attendance will be taken at the beginning of class with a signup sheet. The responsibility lies with the student to sign the sheet. Excuses can be given for medical issues with a doctor's note or some major events, subject to professor's discretion.

Tentative schedule (subject to changes)

Week	Date(s)	Topic	Assignment Due
1	13, 15 January	Course Overview Introduction to Base Python •Arithmetic Operators •Boolean Operators •Data types •Data Structures •Control Flows	
2	20, 22 January	Introduction to Base R; Functions in Python & R •Arithmetic Operators •Boolean Operators •Data types	Assignment 1 Due 20 Jan

		•Data Structures •Control Flows •Functions (R & Python)	
3	27, 29 January	DataFrames! •Introduction to Pandas •Introduction to the Tidyverse •Reading in data •Writing Data •Exploratory Data Analysis (EDA) •Manipulating Data •wide v. long •Group by •Complex slicing and indexing •Loc, iloc, boolean filtering •Data Types •Pandas functions •Tidyverse functions •Merging: Left, Outer, Inner	Assignment 2 Due 27 Jan
4	3,5 February	Application with All Day Exercise in Teams; In the Eyes of an Angel	Assignment 3 Due 3 Feb Assignment 4 (at end of 5 Feb class)
5	10, 12 February	Introduction to Visualization •Understanding & recognizing data types •Types: Ordinal, Nominal, Continuous, Discrete •How to choose the right visualization (based on data type) •Matplotlib •ggplot2	
6	17, 19 February	Bivariate Statistical Analysis •How to choose a statistical test (based on data type) •T-Test •ANOVA •Correlation	Semester Long Project Deliverable #1 Due 17 Feb

			Assignment 5 Due 19 Feb
7	24 Feb	Alternatives to Pandas •How to deal with larger-than-memory data •Dask •Polars	
	26 Feb	Midterm Review	Assignment 6 Due 26 Feb
8	3, 5 March	Supervised Learning •Supervised v. Unsupervised Learning •Linear Models and Non-Linear Models •Sklearn •Classification v. Regression Problems •Classification Models •Logistic Regression •Decision Tree (Classification) •Regression Models •Linear Regression •Decision Tree (Regression) •Machine learning workflow •Train test split •Dealing with categorical features •Dummy variables •Label encoding •Evaluating a model •Classification •Classification Evaluation: Precision, Recall, Confusion matrix, Accuracy, ROC curve, Area under the curve •Regression •Evaluation of Regression: R-squared, MAE, MAPE, RMSE	
9	17, 19 March	Unsupervised learning •Clustering: Exclusive, Agglomerative, Overlapping, Hierarchical •K-means deep dive •Understanding how it works •Understanding use cases and application •Preprocessing steps •PCA •Understanding application	Semester Long Project Deliverable #2 Due 17 March

		•Preprocessing steps	
10	24, 26 March	Application with All Day Exercise in Teams; Lotus Hotel Group	Assignment 7 Due 24 March
11	31 March, 2 April	Dealing with unstructured text data •Understanding challenges in text data •Understanding NLP applications •Text analysis pipeline, common preprocessing steps •Word counts & tf-idf •Applying to classification problem	Assignment 8 Due 31 March Assignment 9 (end of class on 2 April)
12	7, 9 April	Ancillary technology •Introduction to command line interfaces •Terminal •CMD •Basic commands •Git & Github •Understanding why we need Git •Understanding the Git workflow •Basic commands •Init, add, commit, push, pull, status	Semester Long Project Deliverable #3 Due 7 April
13	14 April	Final Exam Review	Assignment 10 Due 14 April
	16 April	Semester Long Project Presentations	
14	21 April	Semester Long Project Presentations	Semester Long Project Deliverable #4
	23 April	Final Exam	

Grading, Exams, Attendance

Grading Framework

Evaluation Component	Weight %	Notes
Assignments	20	Cumulatively – 10 total assignments
Class Attendance	10	Minimum 60% attendance and participation required to pass
Semester Long Project Deliverable #1	4	
Semester Long Project Deliverable #2	12	
Semester Long Project Deliverable #3	12	
Semester Long Project Deliverable #4	12	
Midterm Exam	13	
Final Exam	17	
Total	100	

Exams

Midterm will be conducted in class and format will be discussed during review session.
Final Exam date and other details will be confirmed by undergraduate scheduling office.
Format will be discussed during review session.

Grading Scale

A	94-100
A-	90-93.9
B+	87-89.9
B	84-86.9
B-	80-83.9
C+	77-79.9
C	74-76.9
C-	70-73.9
D+	67-69.9
D	64-66.9
D-	60-63.9
Fail:	<60

University Policies and Resources

University policies:

University policy on observance of religious holidays

In accordance with University policy, students should notify faculty during the first week of the semester of their intention to be absent from class on their day(s) of religious observance. For details and policy, see: students.gwu.edu/accommodations-religious-holidays.

Academic integrity code

Academic dishonesty is defined as cheating of any kind, including misrepresenting one's own work, taking credit for the work of others without crediting them and without appropriate authorization, and the fabrication of information. For details and complete code, see: studentconduct.gwu.edu/code-academic-integrity

Safety and security

In the case of an emergency, if at all possible, the class should shelter in place. If the building that the class is in is affected, follow the evacuation procedures for the building. After evacuation, seek shelter at a predetermined rendezvous location.

Support for students outside the classroom

Disability Support Services (DSS)

Any student who may need an accommodation based on the potential impact of a disability should contact the Disability Support Services office at 202-994-8250 in the Rome Hall, Suite 102, to establish eligibility and to coordinate reasonable accommodations. For additional information see: disabilitysupport.gwu.edu/

Mental Health Services 202-994-5300

The University's Mental Health Services offers 24/7 assistance and referral to address students' personal, social, career, and study skills problems. Services for students include: crisis and emergency mental health consultations confidential assessment, counseling services (individual and small group), and referrals. For additional information see: counselingcenter.gwu.edu/

GW Campus Emergency Information

GW Emergency Services: 202-994-6111

For situation-specific instructions, refer to [GW's Emergency Procedures guide](#). s

GW Alert

GW Alert is an emergency notification system that sends alerts to the GW community. GW requests students, faculty, and staff maintain current contact information by logging on to alert.gwu.edu. Alerts are sent via email, text, social media, and other means, including the Guardian app. The Guardian app is a safety app that allows you to communicate quickly with GW Emergency Services, 911, and other resources. Learn more at safety.gwu.edu.

Protective Actions

GW prescribes four protective actions that can be issued by university officials depending on the type of emergency. All GW community members are expected to follow directions according to the specified protective action. The protective actions are Shelter, Evacuate, Secure, and Lockdown (details below). Learn more at safety.gwu.edu/gw-standard-emergency-statuses.

Shelter

- Protection from a specific hazard
- The hazard could be a tornado, earthquake, hazardous material spill, or other environmental emergency.
- Specific safety guidance will be shared on a case-by-case basis.

Action:

- Follow safety guidance for the hazard.

Evacuate

- Need to move people from one location to another.
- Students and staff should be prepared to follow specific instructions given by first responders and University officials.

Action:

- Evacuate to a designated location.
- Leave belongings behind.
- Follow additional instructions from first responders.

Secure

- Threat or hazard outside of buildings or around campus.
- Increased security, secured building perimeter, increased situational awareness, and restricted access to entry doors.

Action:

- Go inside and stay inside.
- Activities inside may continue.

Lockdown

- Threat or hazard with the potential to impact individuals inside buildings.
- Room-based protocol that requires locking interior doors, turning off lights, and staying out of sight of corridor window.

Action:

- Locks, lights, out of sight
- Consider Run, Hide, Fight